MIL-R-214E

20 September 1976

Superseding
MIL-R-214D
4 October 1966

MILITARY SPECIFICATION

REEL, PHOTOGRAPHIC FILM; AND CAN OR CASE, PHOTOGRAPHIC FILM STORAGE (16MM AND 35MM FILM)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers reels and cans or cases for both 16mm and 35mm motion picture film.

1.2 Classification. Reels and cans or cases shall be of the following standard sizes, as specified (see 6.2).

Type I - 16mm 200,400,800,1200,1600,2000 foot capacity
Type II - 35mm 1000, 2000 foot capacity

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet Materials-Handling, Wood (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall

FSC 6760

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PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock, (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Non-Metallic (And Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	Tape, Pressure-Sensitive, Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-17667	Paper, Wrapping, Chemically Neutral (Non Corrosive)
MIL-W-12332	Welding, Resistance, Spot, Seam, and Projection, for Fabricating Assemblies of Low-Carbon Steel
MIL-F-14072	Finish For Ground Signal Equipment

STANDARDS

FEDERAL

FED-STD-356	Commercial Packaging of Supplies and Equipment

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810	Environmental Test Methods

(Copies of Specifications, Standards, Drawings and Publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.1.

3.2 Material. Unless otherwise specified in the invitation for bid and contract, reels, and cans shall be fabricated from steel. If other materials are used, the same requirements will apply unless excepted in the invitation for bid and contract.

3.3 Reels. Reels shall consist of a hub and two flanges, assembled by welding or riveting for steel or by a method applicable to the type of material specified (see 3.2).

3.3.1 Flanges. Surfaces and the edges of the flanges shall be smooth and free from burrs and sharp edges.

3.3.2 Hubs. Reel hubs shall be securely joined to reel flanges, so as to make a rigid assembly and to insure the proper spacing of the flanges for the respective 16mm or 35mm film. In no case shall there be any projections or rough spots which might cause injury to an operator or film.

3.3.3 Assembly.

3.3.3.1 Riveting. Rivets shall conform to requirement 12 of MIL-STD-454. Rivets or other fastening members of 16mm reels shall not extend beyond the outside surfaces of the flanges more than 1/32 inch. Rivets or other fastening members shall not extend beyond the overall thickness of the reel of .962 inches for 16mm and of 1.885 inches for 35mm reels.

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3.3.3.2 Welding. All welding, if used, shall be in accordance with MIL-W-12332 and shall enable the equipment to meet its performance and other requirements as specified herein.

3.3.3.3 Bonding (Glueing). Assembly of flanges to hub for materials other than steel, as specified by the contract or purchase order, shall be accomplished by a method commensurate with the materials used as approved by the Government procuring activity.

3.4 Reel dimensions.

3.4.1 16MM reels. Dimensions for 16mm reels shall conform to the dimensions specified in Figure 1 and Tables I and II.

3.4.2 35MM reels. Dimensions for 35mm reels shall conform to the dimensions specified in Figure 2 and Table III.

3.5 Flange openings. The flanges of reels shall be provided with means to permit easy access to the hub periphery through openings in the flanges. Four or more finger holes shall be provided in the hub area of the flanges for reels having a film capacity of 800 feet or more.

3.6 Hub slots. Hubs shall have two or more peripheral slots into which the end of the film can be inserted, except that on the 200 and 400 foot 16mm and 1000 foot 35mm reels, only one slot is necessary. The length of the slots shall be the full hub width between the flanges and the slot width shall be at least 0.031 inch. In order to provide easy access to the slots, they shall be located along the center line of a reel flange opening.

3.7 Reel spindle hole.

3.7.1 16MM. The 16mm spindle hole shall be in accordance with Figure 1 and Table II.

3.7.2 35MM. The 35mm reel spindle hole shall be in accordance with Figure 2 and Table III.

3.8 Static spread. When the two flanges are spread apart at adjacent points on their peripheries, as shown in Figure 3, by applying the force and maximum spread distance specified in Table IV, fastenings shall not loosen or fail, nor shall the finish crack, chip or flake. After the force is released, the separation of the two flanges shall be within 0.045 inch of the separation at the same point immediately prior to the application of the force. There shall be no permanent deformation.

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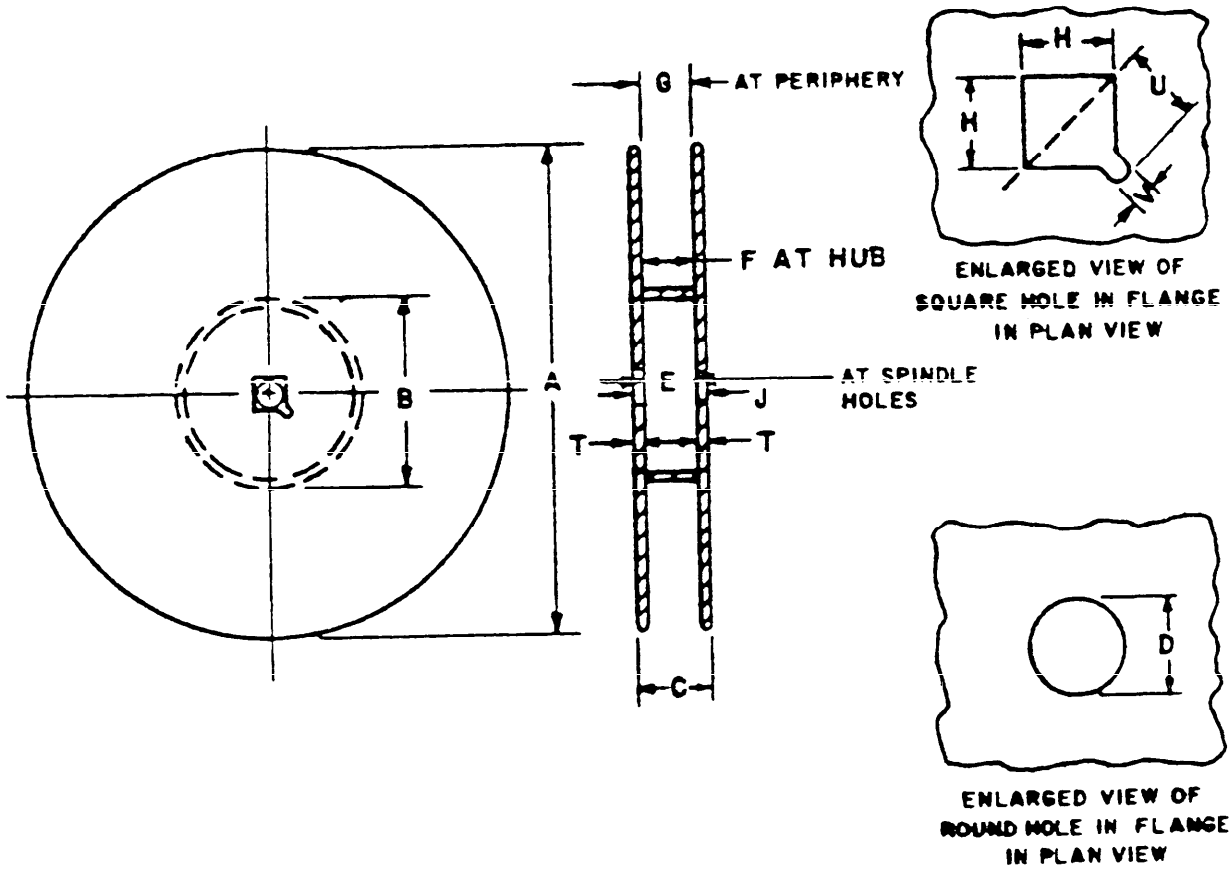


FIGURE 1. 16mm FILM REEL
(FIGURE SHOWS PLAN VIEW AND CROSS SECTION OF REEL)

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Table I. (16MM Reel Dimensions)

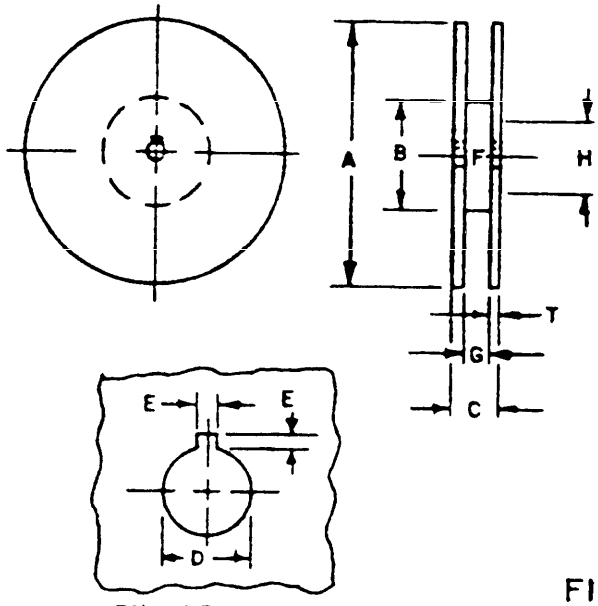
Nominal Capacity	Dimension A				Dimension B				Lateral Run Out	
	Inches		Millimeters		Inches		Millimeters		Inches	MI
200 ft (60 m)	5.031	+0.000 -0.031	127.79	+0.00 -0.79	1.750	±0.250	44.45	±6.35	0.057 max	
400 ft (120 m)	7.000	+0.000 -0.031	177.80	+0.00 -0.79	2.500	+0.000 -0.075	63.50	+0.00 -1.90	0.080 max	
800 ft (240 m)	10.500	+0.000 -0.031	266.70	+0.00 -0.79	4.875	+0.000 -0.375	123.82	+0.00 -9.52	0.120 max	
1200 ft (370 m)	12.250	+0.000 -0.125	311.15	+0.00 -3.18	4.875	+0.000 -0.250	123.82	+0.00 -6.35	0.140 max	
1600 ft (490 m)	14.000	+0.000 -0.275	355.60	+0.00 -6.98	4.875	+0.000 -0.250	123.82	+0.00 -6.35	0.160 max	
2000 ft (610 m)	15.000	+0.000 -0.031	381.00	+0.00 -0.79	4.875	+0.000 -0.250	123.82	+0.00 -6.35	0.171 max	

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Table II. (16MM Reel, Cross Section and Plan View of Reel)

	Dimensions	Inches		Millimeters	
C	Total thickness (including flared, rolled or beveled edges, if any)	0.962	max	24.43	max
D	Spindle hole diameter	0.319	+0.000 -0.003	8.10	+0.00 -0.08
E	Distance between flanges at spindle holes	0.660	±0.015	16.76	±0.38
F	At hub	0.660	±0.010	16.76	±0.25
G	At periphery	0.660	+0.075 -0.000	16.76	+1.90 -0.00
H	Side of square spindle hole	0.319	+0.000 -0.003	8.10	+0.00 -0.08
J	Overall thickness at spindle holes	0.790	max	20.07	max
T	Flange thickness (adjacent to spindle holes)	0.105 0.027	max min	2.67 0.69	max min
U	Keyway depth	0.330	+0.000 -0.020	8.38	+0.00 -0.51
V	Keyway width	0.125	+0.005 -0.000	3.18	+0.13 -0.00
Flange and hub eccentricity		0.031	max	0.79	max

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ENLARED VIEW
OF HOLE IN BOTH FLANGES

FIGURE 2
35mm REEL

Nominal Reel Capacity				
Feet	Meters	Dimensions	Inches	Millimeters
1,000	305	A	11.5 ±0.03	292.1 ±0.8
2,000	610	A	15.0 ±0.03	381.0 ±0.8
		B	5.0 ±0.10	127.0 ±0.8
		C	1.885 ±0.03	47.88 ±0.8
		D	0.317 Min	8.05 Min
			0.319 Max	8.10 Max
		E	0.156 ±0.006	3.96 ±0.15
		F	1.625 ±0.03	41.28 ±0.8
		G	$1.530^{+0.075}_{-0.030}$	38.86 ±0.8
		H	2.25 Min	57.2 Min
		T	0.040 - 0.047	1.02 - 1.19

1 defines area over which reel thickness, specified by dimension F, applies.

Table III. 35MM Reel

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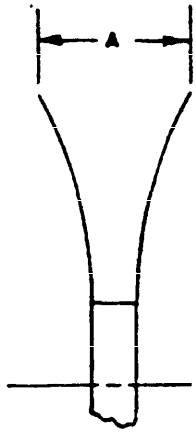


FIGURE 3. SPREADING OF FLANGE

Table IV. Maximum Allowable Spread For Reels

Reel size (feet)	Spreading force (pounds)	Maximum allowable spread at periphery "A" (inches)
16mm Reels		
200	3-1/2	1-1/4
400	5-1/2	1-1/2
800	6-1/2	2-1/2
1,000	8	3
1,600	10	3
2,000	11	3-1/2
35mm Reels		
1,000	8	2-1/4
2,000	10	2-3/4

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3.9 Dynamic spread. When tested in accordance with paragraph 4.9, the welds and rivets shall show no signs of weakening or separation at the fastening points. Following the test, the distance between the flanges, measured at the periphery, shall not exceed one inch for the 16mm reels and one and seven-eights inches for the 35mm reels. Cracking, chipping, or flaking of paint finish shall be cause of rejection of the reels.

3.10 Resistance to squeezing. There shall be no permanent deformation, failure or loosening of fastenings, or chipping or cracking of finish when the reel flanges are squeezed (at a spoke or radial support, if any) in accordance with Figure 4 and Table V.

3.11 Lateral run-out. The maximum excursion of any point on the periphery of a reel flange from the intended plane of rotation of that point when the reel is rotated on an accurate and tightly fitted shaft shall be as follows:

16mm Reels - As specified in Table I
35mm Reels -

1,000 Feet - 0.140 inches maximum
2,000 Feet - 0.171 inches maximum

3.12 Balance. Reels shall have both static and dynamic balance, as far as is consistant with the other requirements of this specification.

3.13 Cans. Cans shall be of two piece construction. The lid and bottom of the can shall fit together snugly. The can shall be embossed or otherwise reinforced at the edges and surfaces to prevent buckling and to prevent identation from other objects laid upon it. All lids shall have a centrally located, flat circular area in order to provide a convenient place for posting labels. The can top and bottom shall be flat to within plus or minus 1/16 inch except for embossing. The embossing of the lid shall be outward; the embossing of the can bottom shall be inward; and the pattern of the embossing shall conform to Figures 5 and 6 and Table VI to facilitate nesting the cans for stacking. All lids and bottoms shall have ridges on the periphery, as indicated in Section A-A of Figures 5 and 6. If hemmed reinforcement is provided at the edge of the lid, the hem shall not extend beyond dimension "B" of Figure 5 and dimension "J" of Figure 6, as indicated in Table V.

3.13.1 Case. When specified in the invitation for bid and contract, a plastic combination film can and shipping case shall be provided.

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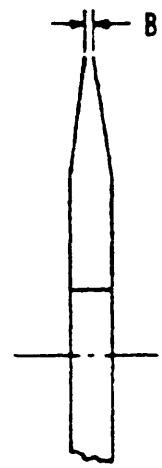
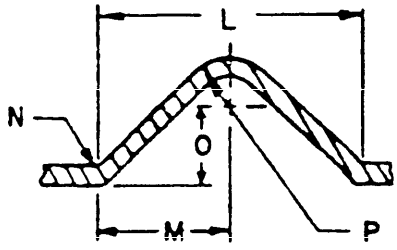


FIGURE 4. SQUEEZING OF FLANGE

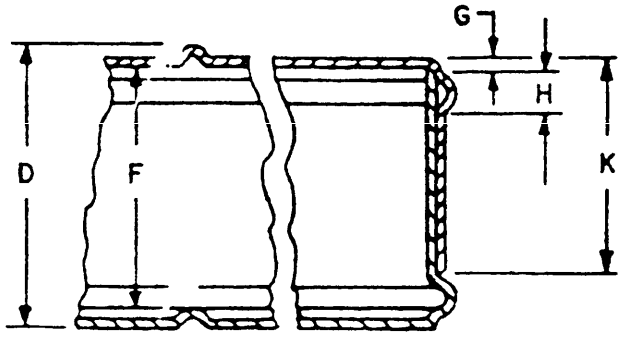
TABLE V DISTANCE BETWEEN FLANGES FOR SQUEEZE TEST

REEL SIZE (FEET)	DISTANCE BETWEEN FLANGES AT PERIPHERY B (INCHES)
16mm REELS	
200	1/4
400	0
800	0
1,200	0
1,600	0
2,000	0
35mm REELS	
1,000	3/4
2,000	3/4

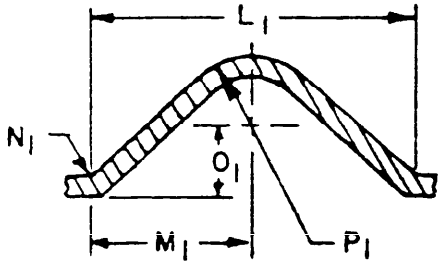
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SECTION B-B
TOP OF CAN EMBOSSING PROFILE



SECTION A-A



SECTION B-B
BOTTOM OF CAN EMBOSSING PROFILE

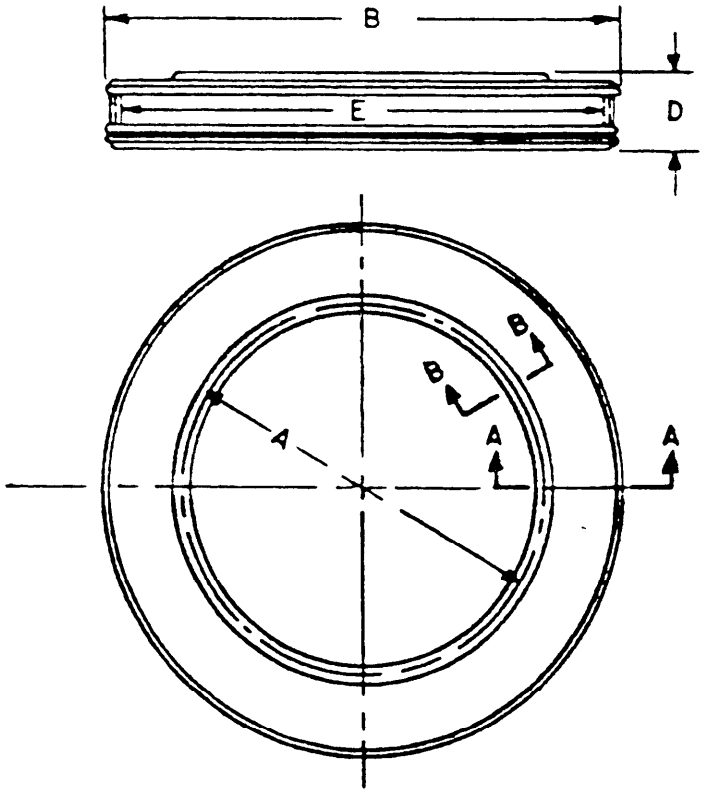
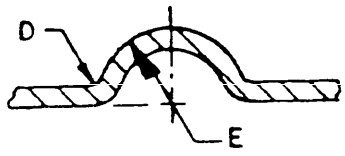
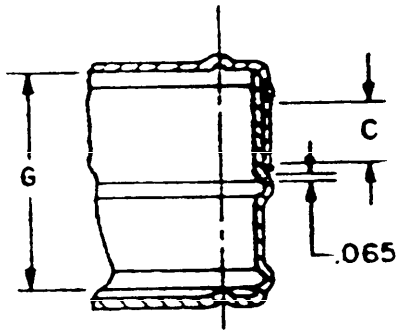


FIGURE 5, 16mm CAN

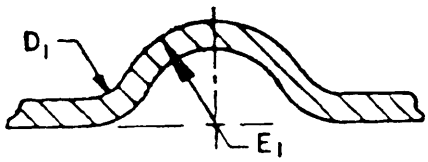
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SECTION B-B
TOP OF CAN EMBOSSING PROFILE



SECTION A-A



SECTION B-B
BOTTOM OF CAN
EMBOSSING PROFILE

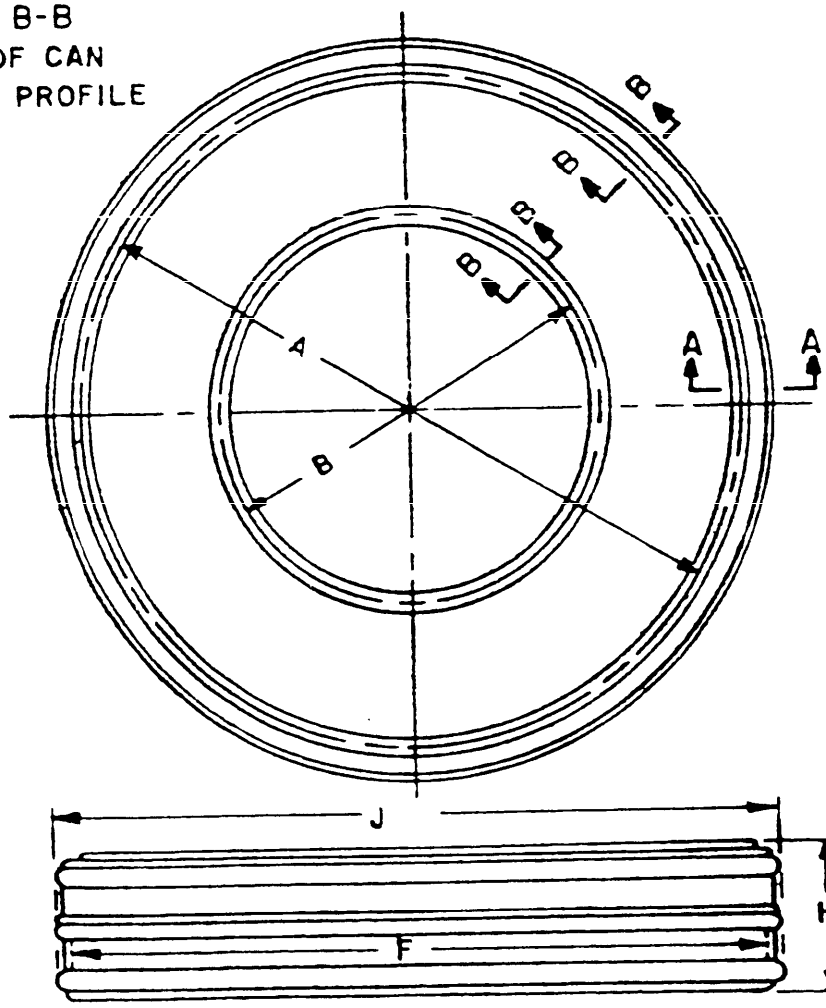


FIGURE 6 35 mm CAN

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Table VI. Can Dimensions

a. 16MM Can						
Dimensions	200 ft cap	400 ft cap	800 ft cap	1,200 ft cap	1,600 ft cap	2,000 ft cap
Embossing pattern						
A--Top and bottom	3.500	5.250	5.250	5.250	5.250	5.250
B--OD including embossing	5.322	7.312	10.812	12.562	14.312	15.312
D--Overall height of can	1.046	1.046	1.046	1.046	1.046	1.046
E--Inside diameter*	5.125 max	7.125 max	10.625 max	12.375 max	14.125 max	15.125 max
	5.062 min	7.062 min	10.562 min	12.312 min	13.725 min	15.062 min
F--Inside depth	0.812 max	0.812 max	0.812 max	0.812 max	0.812 max	0.812 max
	0.781 min	0.781 min	0.781 min	0.781 min	0.781 min	0.781 min
G	0.078	0.078	0.078	0.078	0.078	0.078
H	0.093	0.093	0.093	0.093	0.093	0.093
K	0.734	0.734	0.734	0.734	0.734	0.734
L	0.187	0.187	0.187	0.187	0.187	0.187
L_1	0.234	0.234	0.234	0.234	0.234	0.234
M	0.093	0.093	0.093	0.093	0.093	0.093
M_1	0.109	0.109	0.109	0.109	0.109	0.109
N	0.015	0.015	0.015	0.015	0.015	0.015
N_1	0.031	0.031	0.031	0.031	0.031	0.031
O	0.031	0.031	0.031	0.031	0.031	0.031
O_1	0.015	0.015	0.015	0.015	0.015	0.015
P	0.067	0.067	0.067	0.067	0.067	0.067
P_1	0.084	0.084	0.084	0.084	0.084	0.084
Material	$0.021 \pm .003$	$0.021 \pm .003$	$0.021 \pm .003$	$0.021 \pm .003$	$0.021 \pm .003$	$0.021 \pm .003$

All dimensions listed are in inches

*Inside diameter is measured from the inner opening of the can bottom.

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Table VI. Can Dimensions (Cont)

b. 35MM Can	1,000 ft cap	2,000 ft cap
Embossing pattern and profile		
A	9.375	13.875
B	5.187	5.187
C	0.390	0.390
D	0.015	0.015
D_1	0.031	0.031
E	0.093	0.093
E_1	$0.109^{+.010}_{-.000}$	$0.109^{+.010}_{-.000}$
F-- Inside diameter*	11.625 max 11.595 min	15.125 max 15.095 min
G--Inside depth	1.855 max 1.812 min	1.855 max 1.812 min
H--Overall height of can	2.062	2.062
J--Maximum outside diameter	11.726	15.226
Material	0.021 ±0.003	0.021 ±0.003

*Inside diameter is measured from the inner opening of the can bottom.

NOTES: 1. Tolerances not specified shall be ±.005 inch; all others as indicated.
2. All dimensions listed are in inches.

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3.13.1.1 Construction. The case shall be of a two piece construction. The lid and bottom of the case shall lock together tight so as to prevent any entrance of water when subjected to a rain test (see 4.12.2).

3.13.1.2 Address card holder and address card. The case shall be provided with a holder which shall accommodate a 4 inch by 6-1/2 inch address card. It shall be equipped with an easily operated device to hold the card securely and to provide ready release of the card. Format of the address card shall be as specified in the contract.

3.14 Resistance to salt spray. After being tested as specified in 4.8, empty film reels and open reel cans shall shown no evidence of corrosion.

3.15 Drop. After being tested as specified in paragraph 4.6.1.1 thru 4.6.1.3, the reels shall show no evidence of permanent deformation or failure of or loosening of fastenings. The periphery of the cans shall not be damaged so that the reel cannot be removed. After removal from the can or case, the reel shall shown no evidence of damage to the reel or film.

3.16 Service conditions. Reels, of material other than steel, and cases (see 3.13.1) shall meet the following service conditions.

3.16.1 Temperature.

3.16.1.1 High temperature. After subjection to the test specified in 4.12.3 the reel shall meet the requirements of 3.9.

3.16.1.2 Low temperature. After subjection to the test specified in 4.12.3 the reel shall meet the requirements of 3.9.

3.16.2 Moisture resistance. After subjection to the test specified in 4.12.1 the reel shall meet the requirements of 3.9.

3.16.3 Rain. There shall be no evidence of water inside the case when tested as specified in paragraph 4.12.2.

3.17 Cleaning. After assembly, units shall be cleaned thoroughly and shall be free from particles of flux and other foreign material.

3.18 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072.

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3.19 Marking. All reels and reel cans shall bear identification marking in accordance with the applicable provisions of MIL-STD-130.

3.20 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.3.3.1 Riveting
3.3.3.2 Welding
3.17 Cleaning

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include preparation for delivery) (See 4.3).

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.14).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish six first article units of the complete reel, photographic film; and can or case, photographic film storage (16mm and 35mm film).

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4.3.2 First article inspection. This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for Group A, Group B, and Group C. (See Tables VII, VIII and IX respectively.) The inspection shall be performed in the following order: 4.4, Group A, Group B and Group C for all units. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary document as part of the inspection of equipment required by this specification, and the inspection specified in the contract or purchase order.

Item	Where Required
Finish	3.18

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table VI. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the General Inspection Levels, and AQL's indicated in Table VII.

Table VII. Group A Inspection

Inspection	Req Para	Insp Para	AQL	
			Major	Minor
Visual and Mechanical	3.20	4.13	1%	4%
Lateral Run-Out and Balance	3.11 & 3.12	4.7	1%	4%
Squeezing	3.10	4.11	1%	4%

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4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table VIII and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQL's listed in Table VIII, shall be as follows:

AQL	Inspection Level
15.0 Percent	S-3

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in Table VIII.

Table VIII. Group B Inspection

Inspection	Req Para	Insp Para	AQL
Dynamic Spread	3.9	4.9	15%
Statis Spread	3.8	4.10	15%

4.5.3 Reinspection of conforming Group B sample units. Unless otherwise specified, sample units which have been subject to and passed Group B inspection may be accepted on contract, provided they are resubjected to and pass Group A inspection after repair of all visible damage.

4.5.4 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IX. Samples shall be selected in accordance with 4.5.4.1.

4.5.4.1 Sampling for Group C inspection. One sample reel and can or case shall be selected at random from each 100 units produced for all inspections in Table IX. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

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Table IX. Group C Inspection

Inspection (For Additional First Article) (Inspection See 4.3.2)	Req Para	Insp Para
Temperature	3.16.1	4.12.3
Rain	3.16.3	4.12.2
Moisture Resistance	3.16.2	4.12.1
Salt Spray	3.14	4.8
Drop	3.15	4.6

4.5.4.2 Order of inspection within Group C. Group C inspection shall be performed in accordance with Table IX.

4.5.4.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.5 Disposition of Group C sample units. Sample units which have been subjected to Group C inspection shall not be delivered on this contract.

4.6 Drop test.

4.6.1 Ambient temperature.

4.6.1.1 Reels. Reels loaded to rated capacity, shall be dropped 10 times on a concrete floor from a height of 3 feet. The reels shall be dropped flat alternately on one side and then the other. Reels shall then be examined to determine compliance with paragraph 3.15.

4.6.1.2 Cans. Reels cans, carrying fully loaded film reels, shall be dropped 5 times at random on a concrete floor from a height of 3 feet. The reel shall then be removed from the can and the reel and can examined to determine compliance with paragraph 3.15.

4.6.1.3 Cases. Cases, carrying fully loaded film reels, shall be dropped 5 times at random on a concrete floor from a height of 3 feet. The reel shall then be removed from the case and the reel and case examined to determine compliance with paragraph 3.15.

4.6.2 Low temperature. Cases, carrying fully loaded film reels, shall be dropped 5 times at random on a concrete floor from a height of 3 feet after being stored at -40 degrees F for two hours. The reel shall then be removed from the case and the reel and case examined to determine compliance with paragraph 3.15.

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4.7 Lateral run-out and balance. The reels shall be rotated on an accurate and tightly fitted shaft to determine compliance with the requirements specified in paragraph 3.11 and 3.12.

4.8 Salt spray test. Empty reels and open reel cans shall be exposed to salt spray in accordance with Method 509 of MIL-STD-810, and then examined to determine compliance with the requirement specified in 3.14.

4.9 Dynamic spread test. With the reel rotating at 340 revolutions per minute, a stationary wedge shall be inserted between the flanges of the reel to produce a maximum allowable spread at the periphery as specified in Table III. The design of the wedge shall be such as to prevent any damage to the reel. The reel shall be rotated for 5 minutes with the wedge inserted. Upon completion, the reel shall be examined for compliance with paragraph 3.9.

4.10 Static spread. The reels shall be tested to determine compliance with the requirement specified in 3.8.

4.11 Squeeze test. Reel flanges shall be squeezed and then examined for compliance with paragraph 3.10.

4.12 Service conditions.

4.12.1 Moisture resistance. The reels and cases shall be subjected to MIL-STD-810 Method 507 Procedure II. Step 8 of above does not apply to the reels. At the end of the cycling the reels and cases shall be examined for conformance to the requirement specified in paragraph 3.16.2.

4.12.2 Rain test. The case shall be subjected to Method 506 of MIL-STD-810. At the end of the test the case shall be examined for conformance to the requirement specified in paragraph 3.16.3.

4.12.3 Temperature test.

4.12.3.1 High temperature. The reels and cases shall be subjected to MIL-STD-810, Method 501, Procedure 1, omitting steps 5 and 7. Exposure at the high temperature shall not exceed +160 degrees F for a maximum of 4 hours. The equipment shall then be examined for compliance with 3.16.1.1.

4.12.3.2 Low temperature. The reels and cases shall be subjected to MIL-STD-810, Method 502, Procedure 1, omitting steps 5 and 7. Exposure at the low temperature shall not exceed -80 degrees F for a maximum of 24 hours. The equipment shall then be examined for compliance with 3.16.1.2.

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4.13 Visual and mechanical inspection. The reels, cans, and cases shall be examined for the defects listed in Table X.

Table X. Classification of Visual and Mechanical Defects

Classification	Defect
Major	1. Flange cracked or broken 2. Hub cracked or broken 3. Flange openings omitted 4. Hub slots omitted 5. Reel dimensions not as specified 6. Spindle hole dimensions not as specified 7. Finger holes in hub omitted
Minor	1. Hub slot does not accept film 2. Material scratched 3. Address card holder missing 4. Reel color not as specified 5. Surface and edges of flanges not smooth

4.14 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.2(d)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Reel, Photographic Film; and Can or Case, Photographic Film Storage (16MM and 35MM Film) shall be cleaned in accordance with MIL-P-116 process C-1.

5.1.1.2 Drying. Each Reel, Photographic Film; and Can or Case, Photographic Film Storage (16MM and 35MM Film) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

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5.1.1.4.1 Reel, Photographic Film; and Can or Case, Photographic Film Storage (16MM and 35MM Film). Shall be individually unit packed method III as follows: wrap each unit in paper wrapping material conforming to MIL-P-17667, type I, class I. Secure the wrap with paper tape conforming to PPP-T-45, type III, class A. Place each unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c, style OPF. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservation application and unit packing. Each Reel, Photographic Film; and Can or Case, Photographic Film Storage (16MM and 35MM Film) shall be cleaned, dried, preserved and unit packed as specified in 5.1.

5.1.3 Commercial. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2(d)). Shipping containers for level A or B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Reels, Photographic Film; and Cans or Cases, Photographic Film Storage (16MM and 35MM Film), individually unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

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5.2.2 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A or shrink film.

5.2.3 Level B. A quantity of Reels, Photographic Film; and Cans or Cases, Photographic Film Storage (16MM and 35MM Film), individually unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640, shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3.1 Unitization. Palletization shall be required when: containers specified in 5.2.3 do not require pallet mounting; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.3, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A or shrink film.

5.2.4 Commercial packing. Commercial packing shall be in accordance with FED-STD-356.

5.3 Marking.

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5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping container shall be marked in accordance with FED-STD-356.

5.4 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

6. NOTES

6.1 Intended use. The reels and cans or cases are intended for use in storage, shipping, projection, and editing of motion picture film.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type and size of reels and cans required (see 1.2).

c. Material if other than steel (see 3.2).

d. Level A, level B or commercial preservation and packing (see Section 5).

e. If combination case is required (see 3.13.1).

f. First article pack(s) as follows:

(1) Makeup of pack(s).

(2) Number of each kind of pack to be submitted.

(3) Inspection to be performed thereon.

g. Marking and shipping of samples (see 5.3).

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h. Place of final inspection.

i. Format of address card if required.

j. Whether both reel and reel can or case are required.

k. Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Note: Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CUSTODIANS:

Army-EL
Navy-AS
Air Force-99

Review activities:

Navy-AS
Air Force-99
Other-NS

User activities:
Army-MI

Preparing activity:
Army-EL

Project No 6760-0538

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Gov ment or contractor, involved in the use of the specification in procurement of products for ultimate use by th Department of Defense. This sheet is provided for obtaining information on the use of this specification wh h will insure that suitable products can be procured with a minimum amount of delay and at the least cost. ments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and nd to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authc zation to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-21[illegible]E REEL, PHOTOGRAPHIC FILM: A[illegible]) CAN OR CASE, PHOTOGRAPHIC FILM STORAGE (16MM AND 35MM FILM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMB R

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQU ED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO R ID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this s ification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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